Tiny Transformers for Text Classification: A Synthetic Sample Paper. Abstract: This synthetic paper was created to demonstrate ResearchPilot AI with realistic content. We study whether small transformer models can classify news topics accurately. Research question: How well can a compact transformer encoder classify news topics compared to simpler baseline methods?

Methodology: We train a six layer transformer encoder with self attention on the AG News topic classification dataset, using 120000 training examples and 7600 test examples. We compare against a logistic regression baseline and a bag of words model. All experiments run on a single consumer GPU over three random seeds.

Findings: The compact transformer reaches 87.4 percent accuracy on the AG News test set, compared to 82.1 percent for logistic regression and 79.6 percent for the bag of words baseline. Training completes in under two hours. Limitations: The study only covers English news text, the model is not evaluated on longer documents, and the dataset is limited to four topic classes.